

Editorial Office

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Submission: Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

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Editorial Summary

We present an open-source simulation framework for organic optoelectronic compute-in-memory inference with modern vision transformers. Our central contribution, Ensemble HAT, resamples device-to-device mismatch masks every epoch. On Tiny-ViT-5M, it raises fresh-instance accuracy from chance level (10.00%) to 86.37% (mean $\pm$ 1.54%), enabling robust deployment without per-device calibration.

We rigorously falsify three intuitive strategies for breaking the severe nonlinearity ceiling—joint linearization training, all-linear ablations, and MLP-only controls—all converging to an approximately 30% fresh-instance limit. Rather than weakening the study, this negative result exposes a structural generalization barrier in the attention pathway that first-order behavioral surrogates cannot circumvent. An $N = 30$ statistical test (Hartigan’s dip, $p = 0.98$) on the best second-order-surrogate recipe does not reject the unimodal null hypothesis, strengthening the structural-limit claim against alternative multi-basin hypotheses.

Ablation Coverage Note. We report the completed portion of the ablation space and disclose the following gaps: (a) K4 second-order $\alpha \in \{0.75, 1.00\}$ were not evaluated; $\alpha \in \{0.00, 0.25, 0.50\}$ cover the critical range and already establish non-monotonic behavior with peak at $\alpha = 0.25$ ($44.29 \pm 13.78\%$, still 25 pp below deployment threshold). (b) J2–J4 non-ideality probes (heavy-tailed D2D, temperature drift, IR drop) are reported as scalar sanity checks; their per-instance JSON and full training logs are memo-level. (c) The $N = 30$ K2 evaluation used the literal $\delta g_{\text{eff}} = 0.0$ configuration consistent with J1d training; a later eval-script default changed to -1.0 but does not apply to this evaluation chain.

Novelty Claims

- We introduce the first open-source framework for organic optoelectronic CIM with vision transformers, providing PyTorch-native device-profile substitution and hybrid analog/digital deployment.
- We identify hardware-instance overfitting in organic CIM inference and propose Ensemble HAT, a training-time mismatch-mask resampling fix that restores generalization.
- We falsify three severe-nonlinearity mitigation strategies, establishing an approximately 30% structural ceiling that defines the boundary of first-order behavioral simulation.

Audience

Nature Communications readers will find a reusable methodology for evaluating organic device assumptions on vision tasks and a rigorous negative-result anchor that delineates the limits of first-order behavioral simulation.

Related Submissions

This work is not under consideration elsewhere.

Suggested Reviewers

[To be provided]

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Sincerely,

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